Year 7

REVISION PACK

Unit 4

Name: _____

Class: _____

Unit 4 · Fractions of amounts

Worked example: simplifying and fractions of amounts

(a) Write $\frac{20}{32}$ in its simplest form. (b) Work out $\frac{2}{7}$ of \$84 (c) Work out $\frac{2}{3} \times 27$ (d) Work out $18 \times \frac{5}{6}$

(a)	**Step 1**	HCF of 20 and 32	4
	Step 2	Divide top and bottom by 4	$\frac{20}{32} = \frac{5}{8}$
(b)	**Step 1**	Divide by the bottom	$84 \div 7 = 12$
	Step 2	Times by the top	$12 \times 2 = 24$
	Step 3	Answer	$\$24$
(c)	**Step 1**	Divide by the bottom	$27 \div 3 = 9$
	Step 2	Times by the top	$9 \times 2 = 18$
	Step 3	Answer	18
(d)	**Step 1**	Divide by the bottom	$18 \div 6 = 3$
	Step 2	Times by the top	$3 \times 5 = 15$
	Step 3	Answer	15

Practice: simplifying and fractions of amounts

1.1 (a) Write $\frac{15}{25}$ in its simplest form. (b) Work out $\frac{3}{5}$ of \$145 (c) Work out $\frac{3}{4} \times 48$ (d) Work out $21 \times \frac{5}{7}$

1.2 (a) Write $\frac{16}{24}$ in its simplest form. (b) Work out $\frac{4}{9}$ of \$117 (c) Work out $\frac{2}{5} \times 35$ (d) Work out $24 \times \frac{3}{8}$

1.3 (a) Write $\frac{12}{42}$ in its simplest form. (b) Work out $\frac{5}{6}$ of \$162 (c) Work out $\frac{4}{7} \times 63$ (d) Work out $30 \times \frac{2}{5}$

1.4 (a) Write $\frac{27}{36}$ in its simplest form. (b) Work out $\frac{3}{8}$ of \$216 (c) Work out $\frac{5}{9} \times 72$ (d) Work out $36 \times \frac{7}{12}$

1.5 (a) Write $\frac{14}{49}$ in its simplest form. (b) Work out $\frac{7}{10}$ of \$230 (c) Work out $\frac{3}{8} \times 64$ (d) Work out $45 \times \frac{4}{9}$

1.6 (a) Write $\frac{30}{45}$ in its simplest form. (b) Work out $\frac{2}{3}$ of \$138 (c) Work out $\frac{5}{6} \times 54$ (d) Work out $40 \times \frac{3}{10}$

Unit 4 · Fractions and decimals

Worked example: fractions as decimals and mixed numbers

Use $\frac{1}{4} = 0.25$ to write as decimals: (a) $\frac{3}{4}$ (b) $\frac{1}{8}$

Write as mixed numbers: (c) $\frac{9}{4}$ (d) $\frac{17}{5}$

(a)	**Step 1**	3 quarters: times by 3	$3 \times 0.25 = 0.75$
(b)	**Step 1**	An eighth is half a quarter	$0.25 \div 2 = 0.125$
(c)	**Step 1**	Top ÷ bottom	$9 \div 4 = 2 \text{ r } 1$
	Step 2	Answer	$2\frac{1}{4}$
(d)	**Step 1**	Top ÷ bottom	$17 \div 5 = 3 \text{ r } 2$
	Step 2	Answer	$3\frac{2}{5}$

Practice: fractions as decimals and mixed numbers

2.1 Use $\frac{1}{5} = 0.2$ to write as decimals: (a) $\frac{3}{5}$ (b) $\frac{1}{10}$

Write as mixed numbers: (c) $\frac{11}{4}$ (d) $\frac{19}{6}$

2.2 Use $\frac{1}{8} = 0.125$ to write as decimals: (a) $\frac{5}{8}$ (b) $\frac{1}{16}$

Write as mixed numbers: (c) $\frac{13}{3}$ (d) $\frac{23}{5}$

2.3 Use $\frac{1}{20} = 0.05$ to write as decimals: (a) $\frac{7}{20}$ (b) $\frac{1}{40}$

Write as mixed numbers: (c) $\frac{15}{2}$ (d) $\frac{29}{8}$

2.4 Use $\frac{1}{10} = 0.1$ to write as decimals: (a) $\frac{9}{10}$ (b) $\frac{1}{20}$

Write as mixed numbers: (c) $\frac{17}{6}$ (d) $\frac{22}{7}$

2.5 Use $\frac{1}{25} = 0.04$ to write as decimals: (a) $\frac{3}{25}$ (b) $\frac{1}{50}$

Write as mixed numbers: (c) $\frac{31}{9}$ (d) $\frac{27}{4}$

2.6 Use $\frac{1}{2} = 0.5$ to write as decimals: (a) $\frac{3}{2}$ (b) $\frac{1}{4}$

Write as mixed numbers: (c) $\frac{25}{3}$ (d) $\frac{37}{10}$

Worked example: completing a fraction, decimal and percentage table

Complete the table. Write each fraction in its simplest form. Given: 35%, 140%, $\frac{3}{5}$

Step 1	35%: out of 100, then simplify	$35\% = \frac{35}{100} = \frac{7}{20}, \ 0.35$
Step 2	140%: more than 1	$140\% = 1.4 = 1\frac{40}{100} = 1\frac{2}{5}$
Step 3	$\frac{3}{5}$: make the bottom 100	$\frac{3}{5} = \frac{60}{100} = 0.6 = 60\%$

Practice: completing a fraction, decimal and percentage table

3.1 Complete the table. Write each fraction in its simplest form.

Fraction	Decimal	Percentage
		75%
		160%
$\frac{2}{5}$		

3.2 Complete the table. Write each fraction in its simplest form.

Fraction	Decimal	Percentage
		45%
		120%
$\frac{3}{4}$		

3.3 Complete the table. Write each fraction in its simplest form.

Fraction	Decimal	Percentage
		80%
		250%
$\frac{7}{20}$		

Practice: completing a fraction, decimal and percentage table (continued)

3.4 Complete the table. Write each fraction in its simplest form.

Fraction	Decimal	Percentage
		15%
		130%
$\frac{9}{10}$		

3.5 Complete the table. Write each fraction in its simplest form.

Fraction	Decimal	Percentage
		65%
		175%
$\frac{1}{4}$		

3.6 Complete the table. Write each fraction in its simplest form.

Fraction	Decimal	Percentage
		5%
		110%
$\frac{3}{25}$		

Unit 4 · Multiplying and comparing fractions

Worked example: multiplying and comparing fractions

(a) A puppy eats $\frac{3}{4}$ of a packet every day. How many packets in 5 days?

(b) Which is larger, $\frac{5}{6}$ or $\frac{3}{4}$? Show your reasoning.

(c) Show that $\frac{2}{5} \times \frac{5}{8} = \frac{1}{4}$

(a)	**Step 1**	Times by 5	$\frac{3}{4} \times 5 = \frac{15}{4}$
	Step 2	Answer	$3\frac{3}{4}$
(b)	**Step 1**	Make the bottoms the same	$\frac{5}{6} = \frac{10}{12}$, $\frac{3}{4} = \frac{9}{12}$
	Step 2	Answer	$\frac{5}{6}$
(c)	**Step 1**	Top × top, bottom × bottom	$\frac{10}{40}$
	Step 2	Divide top and bottom by 10	$\frac{10}{40} = \frac{1}{4}$

Practice: multiplying and comparing fractions

4.1 (a) A cat eats $\frac{2}{3}$ of a packet every day. How many packets in 7 days?

(b) Which is larger, $\frac{3}{5}$ or $\frac{2}{3}$? Show your reasoning.

(c) Show that $\frac{3}{4} \times \frac{4}{9} = \frac{1}{3}$

4.2 (a) A dog eats $\frac{3}{5}$ of a packet every day. How many packets in 4 days?

(b) Which is larger, $\frac{4}{5}$ or $\frac{7}{10}$? Show your reasoning.

(c) Show that $\frac{2}{3} \times \frac{3}{8} = \frac{1}{4}$

4.3 (a) A rabbit eats $\frac{1}{6}$ of a packet every day. How many packets in 9 days?

(b) Which is larger, $\frac{5}{8}$ or $\frac{2}{3}$? Show your reasoning.

(c) Show that $\frac{5}{6} \times \frac{3}{10} = \frac{1}{4}$

4.4 (a) A horse eats $\frac{4}{7}$ of a packet every day. How many packets in 5 days?

(b) Which is larger, $\frac{3}{4}$ or $\frac{7}{9}$? Show your reasoning.

(c) Show that $\frac{4}{5} \times \frac{5}{12} = \frac{1}{3}$

4.5 (a) A parrot eats $\frac{2}{9}$ of a packet every day. How many packets in 6 days?

(b) Which is larger, $\frac{5}{7}$ or $\frac{2}{3}$? Show your reasoning.

(c) Show that $\frac{3}{7} \times \frac{7}{9} = \frac{1}{3}$

4.6 (a) A goat eats $\frac{3}{8}$ of a packet every day. How many packets in 10 days?

(b) Which is larger, $\frac{7}{12}$ or $\frac{3}{5}$? Show your reasoning.

(c) Show that $\frac{6}{7} \times \frac{7}{12} = \frac{1}{2}$

Unit 4 · Adding and subtracting fractions

Worked example: adding and subtracting fractions and mixed numbers

Work out

(a) $\frac{2}{5}+\frac{3}{10}$ (b) $\frac{5}{6}-\frac{1}{3}$ (c) $1\frac{1}{2}+2\frac{2}{3}$ (d) $4\frac{1}{3}-1\frac{3}{4}$

(a)	Step 1	Make the bottoms the same	$\frac{4}{10}+\frac{3}{10}$
	Step 2	Answer	$\frac{7}{10}$
(b)	Step 1	Make the bottoms the same	$\frac{5}{6}-\frac{2}{6}=\frac{3}{6}$
	Step 2	Divide top and bottom by 3	$\frac{1}{2}$
(c)	Step 1	Improper fractions	$\frac{3}{2}+\frac{8}{3}$
	Step 2	Make the bottoms the same	$\frac{9}{6}+\frac{16}{6}=\frac{25}{6}$
	Step 3	Answer	$4\frac{1}{6}$
(d)	Step 1	Improper fractions	$\frac{13}{3}-\frac{7}{4}$
	Step 2	Make the bottoms the same	$\frac{52}{12}-\frac{21}{12}=\frac{31}{12}$
	Step 3	Answer	$2\frac{7}{12}$

Practice: adding and subtracting fractions and mixed numbers

5.1 Work out

(a) $\frac{3}{8}+\frac{1}{4}$ (b) $\frac{11}{12}-\frac{3}{4}$ (c) $4\frac{1}{5}+2\frac{7}{10}$ (d) $6\frac{1}{4}-2\frac{7}{12}$

5.2 Work out

(a) $\frac{1}{6}+\frac{2}{3}$ (b) $\frac{7}{10}-\frac{1}{5}$ (c) $2\frac{3}{4}+1\frac{1}{2}$ (d) $5\frac{1}{3}-3\frac{5}{6}$

5.3 Work out

(a) $\frac{2}{9}+\frac{1}{3}$ (b) $\frac{5}{8}-\frac{1}{4}$ (c) $3\frac{2}{5}+1\frac{3}{10}$ (d) $4\frac{1}{6}-1\frac{2}{3}$

5.4 Work out

(a) $\frac{1}{4}+\frac{3}{8}$ (b) $\frac{9}{10}-\frac{2}{5}$ (c) $1\frac{5}{6}+3\frac{1}{3}$ (d) $7\frac{1}{2}-4\frac{4}{5}$

5.5 Work out

(a) $\frac{3}{10}+\frac{1}{2}$ (b) $\frac{5}{6}-\frac{1}{4}$ (c) $2\frac{1}{4}+3\frac{5}{8}$ (d) $6\frac{2}{5}-2\frac{3}{4}$

5.6 Work out

(a) $\frac{1}{3}+\frac{1}{4}$ (b) $\frac{4}{5}-\frac{1}{3}$ (c) $5\frac{2}{3}+1\frac{1}{6}$ (d) $3\frac{1}{8}-1\frac{1}{2}$

Unit 4 · Fraction problems

Worked example: a fraction of a fraction

(a) A farm has an area of 900 m². $\frac{1}{3}$ of it is fields. $\frac{2}{5}$ of the fields are wheat. Work out the area that is wheat.

(b) $\frac{2}{5}$ of the children in a school are girls. $\frac{3}{4}$ of the boys are right-handed. What fraction of the children are left-handed boys?

(a)	**Step 1**	Fields: ⅓ of 900	$\frac{1}{3} \times 900 = 300$
	Step 2	Wheat: 2/5 of the fields	$\frac{2}{5} \times 300 = 120$
	Step 3	Answer	$120\ \text{m}^2$
(b)	**Step 1**	Boys: 1 − girls	$1 - \frac{2}{5} = \frac{3}{5}$
	Step 2	Left-handed: 1 − right-handed	$1 - \frac{3}{4} = \frac{1}{4}$
	Step 3	Fraction of a fraction: ×	$\frac{1}{4} \times \frac{3}{5} = \frac{3}{20}$

Practice: a fraction of a fraction

6.1

(a) An area of 768,000 km² is shared. $\frac{1}{6}$ of it is woodland. $\frac{3}{4}$ of the woodland is pine trees. Work out the area that is pine trees.

(b) $\frac{3}{5}$ of the children in a school are boys. $\frac{7}{8}$ of the girls are right-handed. What fraction of the children are left-handed girls?

6.2

(a) An area of 4,800 m² is shared. $\frac{1}{4}$ of it is garden. $\frac{2}{3}$ of the garden is grass. Work out the area that is grass.

(b) $\frac{1}{2}$ of the children in a school are girls. $\frac{4}{5}$ of the boys are right-handed. What fraction of the children are left-handed boys?

6.3

(a) An area of 36,000 km² is shared. $\frac{2}{3}$ of it is farmland. $\frac{1}{4}$ of the farmland is rice. Work out the area that is rice.

(b) $\frac{3}{7}$ of the children in a school are boys. $\frac{5}{6}$ of the girls are right-handed. What fraction of the children are left-handed girls?

Practice: a fraction of a fraction (continued)

6.4

(a) An area of 1,500 m² is shared. $\frac{2}{5}$ of it is car park. $\frac{5}{6}$ of the car park is covered. Work out the area that is covered.

(b) $\frac{5}{9}$ of the children in a school are girls. $\frac{2}{3}$ of the boys are right-handed. What fraction of the children are left-handed boys?

6.5

(a) An area of 90,000 km² is shared. $\frac{3}{4}$ of it is desert. $\frac{2}{9}$ of the desert is sand dunes. Work out the area that is sand dunes.

(b) $\frac{1}{3}$ of the children in a school are boys. $\frac{9}{10}$ of the girls are right-handed. What fraction of the children are left-handed girls?

6.6

(a) An area of 6,300 m² is shared. $\frac{5}{7}$ of it is park. $\frac{3}{5}$ of the park is trees. Work out the area that is trees.

(b) $\frac{4}{9}$ of the children in a school are girls. $\frac{3}{4}$ of the boys are right-handed. What fraction of the children are left-handed boys?

Unit 4 · Fraction problems

Worked example: problems with adding and multiplying fractions

(a) Reem has two bottles. One is $\frac{3}{4}$ full and the other is $\frac{1}{2}$ full. How much must Reem drink to be left with $\frac{1}{4}$ of a bottle?

(b) A rectangle is $1\frac{1}{2}$ cm long and $\frac{2}{3}$ cm wide. Sam says the area is $1\frac{1}{3}$ cm². Is Sam correct? Explain.

(a)	Step 1	Add the water	$\frac{3}{4} + \frac{2}{4} = \frac{5}{4}$
	Step 2	Take away what is left	$\frac{5}{4} - \frac{1}{4} = \frac{4}{4}$
	Step 3	Answer	1 bottle
(b)	Step 1	Improper fraction	$1\frac{1}{2} = \frac{3}{2}$
	Step 2	Area = length × width	$\frac{3}{2} \times \frac{2}{3} = \frac{6}{6} = 1$
	Step 3	Answer	No, the area is 1 cm^2

Practice: problems with adding and multiplying fractions

7.1

(a) Reem has two bottles of water. One is $\frac{4}{5}$ full and the other is $\frac{3}{20}$ full. How much must Reem drink to be left with $\frac{1}{2}$ of a bottle in total?

(b) A rectangle is $2\frac{3}{5}$ cm long and $\frac{3}{4}$ cm wide. Zeinab says the area is $3\frac{9}{20}$ cm². Is she correct? Explain.

7.2

(a) Ali has two bottles of water. One is $\frac{2}{3}$ full and the other is $\frac{5}{6}$ full. How much must Ali drink to be left with $\frac{1}{3}$ of a bottle in total?

(b) A rectangle is $1\frac{1}{4}$ cm long and $\frac{2}{5}$ cm wide. Zeinab says the area is $\frac{1}{2}$ cm². Is she correct? Explain.

7.3

(a) Mona has two bottles of water. One is $\frac{7}{8}$ full and the other is $\frac{1}{4}$ full. How much must Mona drink to be left with $\frac{3}{8}$ of a bottle in total?

(b) A rectangle is $3\frac{1}{3}$ cm long and $\frac{3}{5}$ cm wide. Zeinab says the area is $1\frac{4}{5}$ cm². Is she correct? Explain.

Practice: problems with adding and multiplying fractions (continued)

7.4

(a) Omar has two bottles of water. One is $\frac{3}{5}$ full and the other is $\frac{7}{10}$ full. How much must Omar drink to be left with $\frac{1}{2}$ of a bottle in total?

(b) A rectangle is $2\frac{1}{2}$ cm long and $\frac{4}{5}$ cm wide. Zeinab says the area is $2\frac{3}{10}$ cm². Is she correct? Explain.

7.5

(a) Lina has two bottles of water. One is $\frac{5}{6}$ full and the other is $\frac{1}{3}$ full. How much must Lina drink to be left with $\frac{2}{3}$ of a bottle in total?

(b) A rectangle is $1\frac{3}{4}$ cm long and $\frac{2}{7}$ cm wide. Zeinab says the area is $\frac{1}{2}$ cm². Is she correct? Explain.

7.6

(a) Sami has two bottles of water. One is $\frac{9}{10}$ full and the other is $\frac{2}{5}$ full. How much must Sami drink to be left with $\frac{3}{5}$ of a bottle in total?

(b) A rectangle is $4\frac{1}{2}$ cm long and $\frac{2}{3}$ cm wide. Zeinab says the area is $2\frac{2}{3}$ cm². Is she correct? Explain.